

LAKEVIEW UNIVERSITY

Department of Computer Science

CS 201 - DATA STRUCTURES AND ALGORITHMS

Spring 2026 | 3 Credit Hours | CRN: 71042

Instructor:

Prof. Sandra Okafor

Office:

Tech Center 312

Office Hours: Tuesday 1:00 - 3:00 PM, Thursday 10:00 AM - 12:00 PM

Email:

sokafor@lakeviewu.edu

Meeting Times:

Monday, Wednesday, Friday 8:00 - 8:50 AM

Location:

Wieboldt Hall 214

COURSE DESCRIPTION

This course covers the design, analysis, and implementation of fundamental data structures including arrays, linked lists, stacks, queues, trees, heaps, hash tables, and graphs. Students will analyze algorithm efficiency using Big-O notation and implement solutions to standard computational problems. Lab sessions reinforce concepts through programming assignments in Java.

Prerequisites:

CS 101 and CS 102 with a grade of C or better.

REQUIRED MATERIALS

- Textbook:

Introduction to Algorithms, 4th Edition, Cormen et al.

(available on course reserve at the Lakeview Library)

- Java Development Kit 17 or later (free, install before first lab)

- Access to Lakeview course portal for assignments and grades

GRADING

Component	Weight
Programming Assignments	35%
Midterm Exam	25%
Final Exam	30%
Participation	10%

Grading Scale:

A 93 - 100	A- 90 - 92	B+ 87 - 89	B 83 - 86
B- 80 - 82	C+ 77 - 79	C 73 - 76	C- 70 - 72
D 60 - 69	F below 60		

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COURSE SCHEDULE

Week	Dates	Topic
1	Jan 12-16	Course overview, review of OOP, algorithm analysis intro
2	Jan 20-23	Big-O notation, best/worst/average case, examples
3	Jan 26-30	Arrays and dynamic arrays, amortized analysis
4	Feb 2-6	Linked lists: singly, doubly, circular
5	Feb 9-13	Stacks and queues, array and linked implementations
6	Feb 16-20	Recursion review, recursive data structures
7	Feb 23-27	Trees: binary trees, traversals, BSTs
8	[Week 8]	MIDTERM EXAM (date TBD by registrar)

(Note:

Spring break follows. Class resumes in fall schedule.)

... Oct Heaps and priority queues, heap sort

... Oct Hash tables, collision resolution

... **Nov** **Graphs:**

representations, BFS, DFS

... Nov **Shortest paths:**

Dijkstra, Bellman-Ford

... Dec **Sorting algorithms:** merge sort, quicksort, comparison lower bound

... Dec Final exam review

Finals Week FINAL EXAM (date TBD by registrar)

Note:

Specific dates for October through December are finalized
and posted to the course portal.

PROGRAMMING ASSIGNMENTS

PA 1 Due Feb 6 Linked list implementation in Java

PA 2 Due Feb 27 Binary search tree with insert, search, delete

PA 3 Due Oct 16 Hash table with chaining

PA 4 Due Nov 13 Graph traversal: BFS and DFS

PA 5 Due Dec 4 Dijkstra shortest path implementation

Late work:

10% deducted per day, no submissions accepted after 5 days.

ACADEMIC INTEGRITY

All submitted code must be your own work. You may discuss concepts with
classmates but you may not share code. Use of AI tools to generate submitted
code is not permitted and constitutes an academic integrity violation.

Violations are reported to the Dean of Students.

ACCESSIBILITY

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Students needing accommodations should contact the Office of Disability
Services and provide documentation to the instructor within the first two
weeks of class.

Syllabus subject to minor revision. Updates posted to the course portal.

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